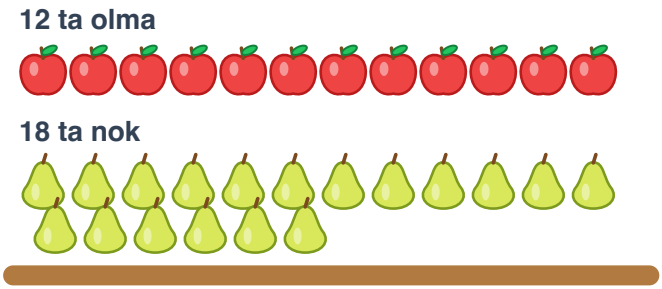
Algebraik ifodalarni ko'paytuvchilarga ajratish

1-QISM

Qiziqtiramiz

SAVOL · SOVG'A SAVATLARI

Bog'bon bobo bayramga 12 ta olma va 18 ta nokni bir xil savatlarga joylamoqchi: har bir savatda olmalar soni bir xil, noklar soni ham bir xil bo'lsin, birorta meva ortib qolmasin.



1-savol. Ko'pi bilan nechta savat tayyorlash mumkin? Har bir savatga nima tushadi?

2-savol. Bitta olma a so'm, bitta nok n so'm tursa, hamma mevalar $12a + 18n$ so'm turadi. Shu summani «savatlar soni × bitta savat narxi» ko'rinishida yozsa bo'ladimi?



Savatlar soni 12 ni ham, 18 ni ham bo'lishi kerak. Ifodani ko'paytma qilib yozishda ham xuddi shu son kerak bo'ladi.

ESLAB OLAYLIK

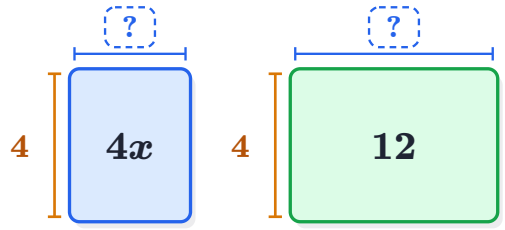
- EKUB — sonlarning eng katta umumiy bo'luvchisi: $\text{EKUB}(18;\ 24) = 6$, chunki $18 = 2 \cdot 3 \cdot 3$, $24 = 2 \cdot 2 \cdot 2 \cdot 3$.
- Qavslarni ochish: $4(2x + 5) = 8x + 20$, $\quad -3(a - 2b) = -3a + 6b$.
- Daraja: $a^2 = a \cdot a$, $\quad 5m^2n = 5 \cdot m \cdot m \cdot n$.

2-QISM

Sinab ko'ramiz

1-TOPSHIRIQ · IKKI TO'G'RI TO'RTBURCHAK

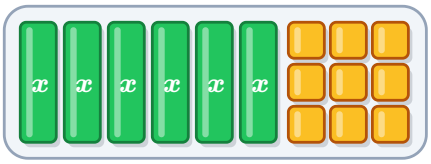
Ikkala to'g'ri to'rtburchakning bo'yi 4 ga teng, yuzlari $4x$ va 12. Ularni yonma-yon qo'yib, bitta katta to'g'ri to'rtburchak yasaymiz.



a) Har birining eni nechaga teng?
b) Katta to'g'ri to'rtburchakning bo'yi va eni qanday?
c) $4x + 12$ ni ikki ifodaning ko'paytmasi qilib yozing.

2-TOPSHIRIQ · PLITKALAR

Yashil uzun plitka — x, sariq kvadratcha — 1. Rasmda $6x + 9$.



$$6x + 9$$

a) Plitkalarni 3 ta teng guruhga ajrating. Har bir guruhda nechta yashil va nechta sariq plitka bor?
b) 2 ta teng guruhga ajratsa bo'ladimi?
c) $6x + 9$ ni ko'paytma qilib yozing.

3-TOPSHIRIQ · KIM HAQ?

Sardor: $12x + 18 = 2(6x + 9)$.
Lola: $12x + 18 = 6(2x + 3)$.

Qavslarni ochib tekshiring: ikkalasi ham to'g'rimi? Kimning qavsi ichida hali umumiy ko'paytuvchi bor?

4-TOPSHIRIQ · TEZ HISOB

Kalkulyatorsiz hisoblang: $47 \cdot 36 + 47 \cdot 64$. Ikkala ko'paytmada qaysi son takrorlanmoqda?

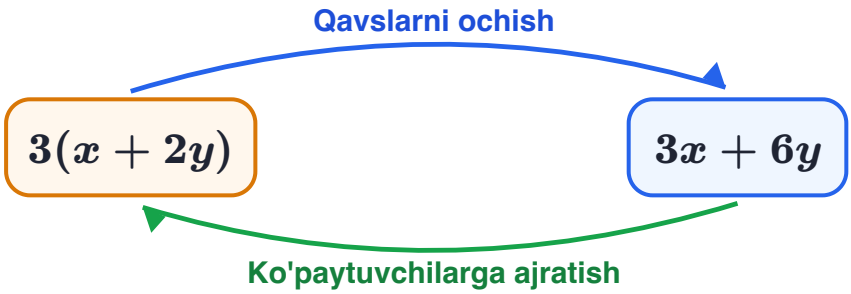
3-QISM

Tushunamiz

TA'RIF · KO'PAYTUVCHILARGA AJRATISH

Algebraik ifodani ko'paytuvchilarining ko'paytmasi ko'rinishida yozish jarayoni **ko'paytuvchilarga ajratish** deyiladi.

Ko'paytuvchilarga ajratish qavslarni ochishga **teskari amaldir**.



Bu yerda $3x+6y$ ifodaning ko'paytuvchilari: 3 va $(x+2y)$.

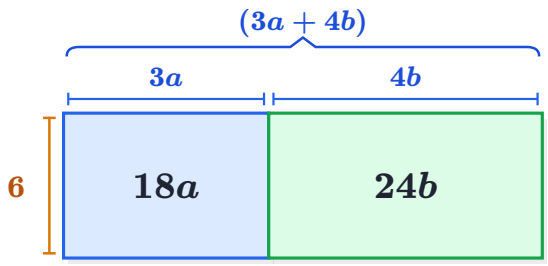
USUL · UMUMIY KO'PAYTUVCHINI QAVSDAN CHIQARISH

1. Hadlarning umumiy ko'paytuvchisini aniqlaymiz: soni — koeffitsiyentlarning EKUBi, harflari — har bir hadda bor harflar.
2. Har bir hadni umumiy ko'paytuvchi va qolgan ko'paytuvchilarning ko'paytmasi ko'rinishida yozamiz.
3. Umumiy ko'paytuvchini qavs tashqarisiga chiqaramiz.

1-MISOL · FAQAT SON UMUMIY

$18a + 24b$. EKUB$(18;\ 24) = 6$, harflar har xil.

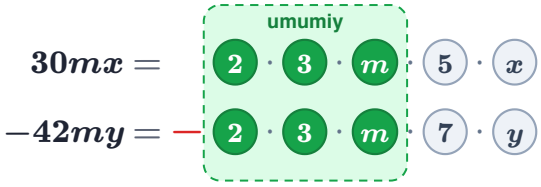
$18a + 24b = 6(3a) + 6(4b) = 6(3a + 4b)$.



$$18a + 24b = 6(3a + 4b)$$

2-MISOL · SON VA HARF UMUMIY

$30mx - 42my$. Har bir hadni ko'paytuvchilarga yoyamiz — ikkalasida bor ko'paytuvchilar yashil:



Umumiy ko'paytuvchi: $6m$

$$30mx - 42my = 6m(5x - 7y)$$

$30mx - 42my = 6m(5x) - 6m(7y) = 6m(5x - 7y)$.

QOIDA · AJRATISH TUGALLANDIMI?

Hadlarning umumiy ko'paytuvchilari qolmasa, ko'paytuvchilarga ajratish tugallangan hisoblanadi.

Ifodani ko'paytuvchilarga ajratish uchun barcha umumiy ko'paytuvchilarni qavsdan tashqariga chiqarishimiz kerak.

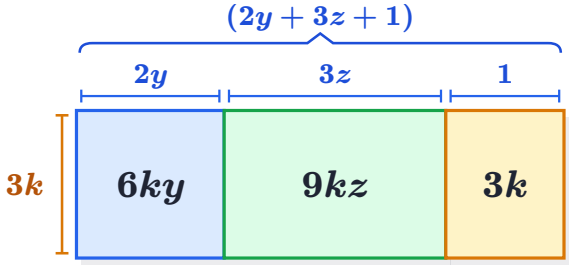
3-MISOL · CHALA QOLGAN AJRATISH

$20pk - 30pm = 10(2pk - 3pm)$ — hali tugamagan: qavs ichidagi ikkala hadda p bor.

p ni ham chiqaramiz: $10(2pk - 3pm) = 10p(2k - 3m)$. Endi $2k$ va $3m$ da umumiy ko'paytuvchi yo'q.

4-MISOL · HAD BUTUNLAY CHIQSA

$6ky + 9kz + 3k = 3k(2y) + 3k(3z) + 3k(1) = 3k(2y + 3z + 1)$.



$$6ky + 9kz + 3k = 3k(2y + 3z + 1)$$

Uchinchi had $3k$ butunlay chiqdi — uning o'rnida qavsda 1 qoladi.

ESLATMA · TEKSHIRISH

Ko'paytuvchilarga ajratish to'g'riligini tekshirish uchun qavslarni ochish va soddalashtirish orqali berilgan ifoda bilan taqqoslash mumkin.

Ko'paytuvchilarga ajratishning har bir bosqichidagi ifodalar o'zaro teng. Masalan, $5x + 15$ va $5(x + 3)$: $x = 2$ da $5 \cdot 2 + 15 = 25$ va $5 \cdot (2 + 3) = 25$.

Qavsni ochganda berilgan ifoda chiqsa — ajratish to'g'ri. Qavs ichida yana umumiy ko'paytuvchi qolmasa — tugallangan.

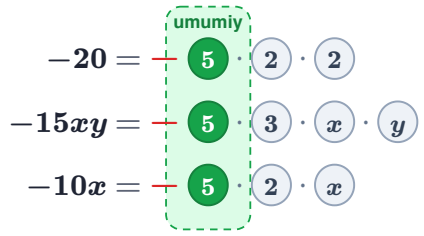


USUL · BIRINCHI HAD MANFIY BO'LSA

Umumiy ko'paytuvchini minus ishorasi bilan chiqaramiz. Shunda qavs ichidagi har bir hadning ishorasi almashadi.

5-MISOL · HAMMA HAD MANFIY

$-20 - 15xy - 10x = -5(4) - 5(3xy) - 5(2x)$
$= -[5(4) + 5(3xy) + 5(2x)] = -5(4 + 3xy + 2x)$.



Umumiy ko'paytuvchi: -5

$-20 - 15xy - 10x = -5(4 + 3xy + 2x)$

6-MISOL · ISHORALAR ARALASH

$-8a + 12b$. Minus bilan chiqaramiz: $-8a + 12b = -4(2a) - 4(-3b) = -4(2a - 3b)$.

Tekshiramiz: $-4 \cdot 2a = -8a$, $-4 \cdot (-3b) = +12b$ ✓. (Minussiz ham to'g'ri: $4(-2a + 3b)$.)

USUL · UMUMIY KO'PAYTUVCHI — QAVS

Ikkala hadda bir xil qavs bo'lsa, shu qavsning o'zi umumiy ko'paytuvchi: uni butunligicha chiqaramiz.

7-MISOL · QAVS CHIQADI

$7k(2a+b)-3m(2a+b)=(2a+b)(7k-3m)$.

$11y(p-q)-4y(p-q)=(p-q)(11y-4y)=(p-q)(7y)=7y(p-q)$.

8-MISOL · SONLAR BILAN

Taqsimot qonuni hisobni ham osonlashtiradi: 73 ikkala ko'paytmada bor.

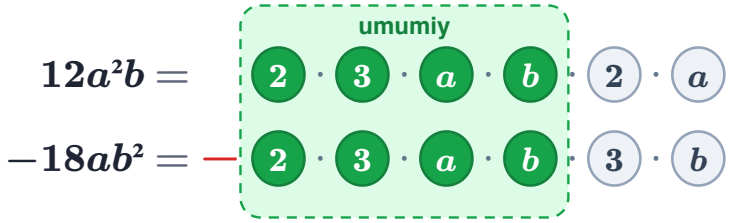
$73\cdot 58+73\cdot 42=73\cdot(58+42)=73\cdot 100=7\,300$.

4-QISM

Qo'llaymiz

1-MISOL · DARAJALI HADLAR

Ko'paytuvchilarga ajrating: $12a^2b-18ab^2$.



Umumiy ko'paytuvchi: $6ab$

$$12a^2b-18ab^2=6ab(2a-3b)$$

1. EKUB$(12;\ 18)=6$. Harflar: a ham, b ham ikkala hadda bor — har biridan bittadan olamiz. Umumiy ko'paytuvchi — $6ab$.
2. $12a^2b=6ab\cdot 2a,\ \ 18ab^2=6ab\cdot 3b$.
3. $12a^2b-18ab^2=6ab(2a-3b)$.

2-MISOL · QAVSLAR QARAMA-QARSHI

$5x(3m-n)-2y(n-3m)$.

1. $n-3m=-(3m-n)$, shuning uchun $-2y(n-3m)=+2y(3m-n)$.
2. $5x(3m-n)+2y(3m-n)=(3m-n)(5x+2y)$.

3-MISOL · AVVAL SODDALASHTIRAMIZ

$3(4x-5)-2(3x-9)$.

1. Qavslarni ochamiz: $12x-15-6x+18$.
2. O'xshash hadlarni ixchamlaymiz: $6x+3$.
3. Ajratamiz: $6x+3=3(2x+1)$.

4-MISOL · JUFTLAB AJRATAMIZ

$(3ab+12a)-(8+2b)$.

1. Har bir qavsni alohida ajratamiz: $3ab+12a=3a(b+4)$, $8+2b=2(4+b)=2(b+4)$.
2. Endi $3a(b+4)-2(b+4)$ — umumiy ko'paytuvchi $(b+4)$.
3. $=(b+4)(3a-2)$.

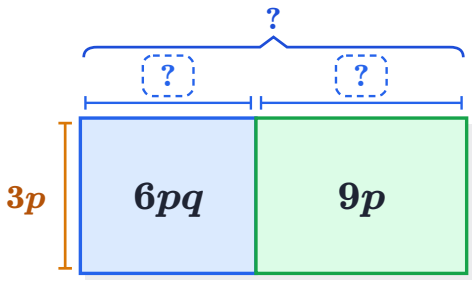
5-MISOL · KALKULYATORSIZ

a) $0{,}64\cdot 37+37\cdot 0{,}36=37\cdot(0{,}64+0{,}36)=37\cdot 1=37$.

b) $58\cdot(-27)-58\cdot 73=58\cdot(-27-73)=58\cdot(-100)=-5\,800$.

6-MISOL · NOMA'LUM TOMON

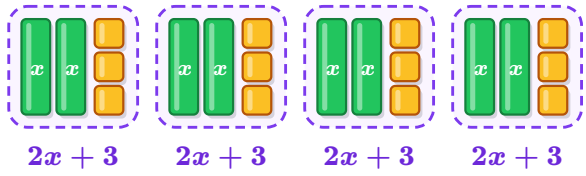
To'g'ri to'rtburchakning yuzi $6pq+9p$, bo'yi $3p$. Enini toping.



$6pq+9p=3p(2q+3)$. Yuz = bo'yi × eni, demak eni $2q+3$.

7-MISOL · PLITKALARNI TENG GURUHLARGA

$8x+12$ ta plitkani bir nechta teng guruhga ajratish kerak. EKUB$(8;\ 12)=4$, demak ko'pi bilan 4 ta guruh:



$$8x+12=4(2x+3)$$

2 ta guruh ham bo'ladi: $8x+12=2(4x+6)$ — lekin bu ajratish tugallanmagan.

8-MISOL · BO'LINISHNI ISBOTLASH

n — natural son. $6n + 6n^2$ har doim 12 ga bo'linishini ko'rsating.

1. $6n + 6n^2 = 6n(1 + n) = 6 \cdot n(n + 1)$.
2. n va $n + 1$ — ketma-ket ikki son, ulardan biri albatta juft. Demak $n(n + 1)$ juft: $n(n + 1) = 2t$.
3. $6 \cdot 2t = 12t$ — 12 ga bo'linadi. Tekshiruv: $n = 3$ da $18 + 54 = 72 = 12 \cdot 6$ ✓.



Had butunlay chiqib ketsa, qavsda 1 qoladi — uni tashlab ketish oson. Men shu joyda ko'p adashardim!

DIQQAT!

1. ✗ $16pq - 24pr = 4(4pq - 6pr)$ ✓ **qavsda hali $2p$ bor:** $8p(2q - 3r)$
2. ✗ $6ky + 9kz + 3k = 3k(2y + 3z)$ ✓ $3k(2y + 3z + 1)$
3. ✗ $-6x - 9 = -3(2x - 3)$ ✓ $-3(2x + 3)$: $-3 \cdot 3 = -9$
4. ✗ $4x + 6xy + 2y = 2x(2 + 3y + y)$ ✓ **$2y$ da x yo'q:** $2(2x + 3xy + y)$
5. ✗ $2m(x + y) + (x + y) = 2m(x + y)$ ✓ **ikkinchi hadda 1 qoladi:** $(x + y)(2m + 1)$

Mashqlar

1. Ko'paytuvchilarga ajrating.
   a) $4m + 10n$ b) $9p - 12q$ c) $15k + 35$ d) $12 + 12t$ e) $22 - 11c$
2. Ko'paytuvchilarga ajrating.
   a) $27kx - 18ky$ b) $-12cm - 4c$ c) $10ab + 5bt$
   d) $7x - 63xy$ e) $-8pt - 20pq$ f) $36mn + 12n$

3. Ko'paytuvchilarga ajrating.
   a) $6x - 10y + 4$ b) $3ab - 12ac + 3a$ c) $-15 - 25t - 5mt$ d) $12mk - 20nk + 28k$

4. Xatoni toping va tuzating.
   a) Bekzod: $18pq - 27pr = 3(6pq - 9pr)$
   b) Malika: $10xy - 15xz + 5x = 5x(2y - 3z)$
   c) Doston: $-9a - 12b = -3(3a - 4b)$

5. Ko'paytuvchilarga ajrating.
   a) $3x(2a + 5) - 4y(2a + 5)$ b) $k(3p - 2) + (3p - 2)$
   c) $5b(m - n) + 13b(m - n)$ d) $8a(c - d) - 12b(c - d)$

6. Ko'paytuvchilarga to'liq ajrating.
   a) $x^2 + 7x$ b) $8a^2b + 12ab$ c) $15m^2n - 10mn^2$
   d) $p^3 + 4p^2q - p^2$ e) $-x^2y^3 - x^3y^2$

7. Kalkulyatorsiz hisoblang.
   a) $56 \cdot 23 + 56 \cdot 77$ b) $4{,}7 \cdot 0{,}38 + 0{,}62 \cdot 4{,}7$ c) $135 \cdot 41 - 35 \cdot 41$
   d) $-6{,}2 \cdot 19 - 19 \cdot 3{,}8$ e) $77 \cdot (-48) + 77 \cdot 148$

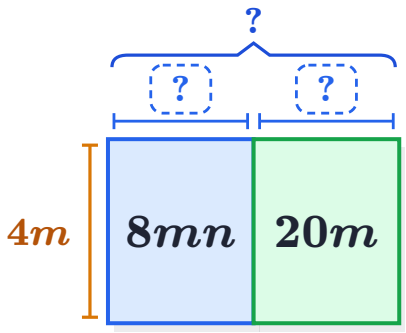
8. Avval ko'paytuvchilarga ajrating, keyin qiymatini toping.
   a) $mk + mt$, bunda $m = 13$, $k = 62$, $t = 38$
   b) $6pk - 18pm$, bunda $p = 4$, $k = 35$, $m = 9$

9. Soddalashtiring va ko'paytuvchilarga ajrating.
   a) $5(3x + 2) - 3(x - 6)$ b) $4(5a - b) - 2(-a + 3b)$

10. Ko'paytuvchilarga ajrating (qavslarga diqqat!).
    a) $4a(2x - y) - 3b(y - 2x)$ b) $(m - 5n) - 3n(-m + 5n)$

11. Juftlab ajrating.
    a) $(5xy + 10x) + (3y + 6)$ b) $(4ab + 12a) - (7b + 21)$

12. a) To'g'ri to'rtburchakning yuzi $8mn + 20m$, bo'yi $4m$. Enini toping.
    b) To'g'ri to'rtburchakning yuzi $7xy - 14x$. Uning tomonlari qanday bo'lishi mumkin? Ikki xil javob yozing.



13. $15x + 10y + 25$ ta plitka bor (yashil — x, binafsha — y, sariq — 1). Ularni 1 tadan ko'p teng guruhlarga ajratishning barcha usullarini toping. Har guruhda nima bo'ladi?

14. Kitob do'koniga 28 quti daftar (har qutida a dona) va 42 quti ruchka (har qutida b dona) keldi. Qutilar ochilmasdan bir nechta maktabga teng bo'linadi: har bir maktabga daftar qutilari ham, ruchka qutilari ham bir xil sonda.
    a) Jami buyumlar sonini ifoda bilan yozing.
    b) Ko'pi bilan nechta maktabga bo'lish mumkin? Ifodani ko'paytuvchilarga ajratib yozing.

15. a) $n^2 + 3n$ ni ko'paytuvchilarga ajrating.
    b) n natural son bo'lganda $n^2 + 3n$ har doim juft ekanini ko'rsating.
    c) $5k + 5k^2$ har doim 10 ga bo'linishini ko'rsating (k — natural son).

16. a) $(3ab + 9a) - (6b + 18)$ ni ko'paytuvchilarga to'liq ajrating.
    b) a ning qanday qiymatida ifoda b ning har qanday qiymatida 0 ga teng bo'ladi?
    c) $a = 5$, $b = 7$ bo'lsa, ifodaning qiymatini toping.

17. A to'g'ri to'rtburchakning tomonlari $5x$ va $2y + 3$, B to'g'ri to'rtburchakniki — $3x$ va $4y + 1$ (x va y musbat).
    a) B ning yuzidan A ning yuzini ayiring va natijani ko'paytuvchilarga ajrating.
    b) y ning qanday qiymatida yuzlar teng? $y = 10$ bo'lsa, qaysi yuz katta?

18. Qaysi ifodalarni $(m - 2n)$ ga ko'paytma qilib yozsa bo'ladi?
    a) $3m - 6n$ b) $2m - n$ c) $am - 2an$ d) $-5m + 10n$ e) $m^2 - 2mn$
    Shunday ifodadan yana bittasini o'zingiz tuzing.

5-QISM

Tekshiramiz

1. Ko'paytuvchilarga ajrating: a) $7a + 14b$ b) $24x - 18$ c) $9mn + 6m$
2. Ko'paytuvchilarga ajrating: a) $45pq - 30pr$ b) $-12y - 20xy$ c) $8ab - 4a + 12ac$
3. Nodira $14xz - 21yz = 7(2xz - 3yz)$ deb yozdi. Ajratish tugallanganmi? Kerak bo'lsa, tuzating.
4. Ko'paytuvchilarga ajrating: $2t(5a - c) + 9t(5a - c)$.
5. Ko'paytuvchilarga to'liq ajrating: $18m^2n + 27mn^2$.
6. Kalkulyatorsiz hisoblang: a) $93 \cdot 64 + 93 \cdot 36$ b) $2{,}5 \cdot 7{,}3 - 2{,}5 \cdot 3{,}3$
7. Soddalashtiring va ko'paytuvchilarga ajrating: $6(2x - 1) - 4(x - 3)$.
8. To'g'ri to'rtburchakning yuzi $12ab + 30a$, bo'yi $6a$. Enini toping.
9. Ko'paytuvchilarga ajrating: $3a(m - 4) - 5b(4 - m)$.
10. Juftlab ajrating: $(6xy + 18x) + (5y + 15)$.

- [ ] Men hadlarning umumiy ko'paytuvchisini (EKUB va umumiy harflar) topa olaman.
- [ ] Men umumiy ko'paytuvchini qavsdan chiqarib, ajratish tugallanganini va to'g'riligini tekshira olaman.
- [ ] Men manfiy ishorani va qavsni umumiy ko'paytuvchi sifatida chiqara olaman.
- [ ] Men ko'paytuvchilarga ajratish bilan hisobni osonlashtira va masala yecha olaman.

Javoblar

4-qism · Mashqlar

1-masala

a) $2(2m+5n)$ b) $3(3p-4q)$ c) $5(3k+7)$ d) $12(1+t)$ e) $11(2-c)$.

2-masala

a) $9k(3x-2y)$ b) $-4c(3m+1)$ c) $5b(2a+t)$ d) $7x(1-9y)$ e) $-4p(2t+5q)$ f) $12n(3m+1)$.

3-masala

a) $2(3x-5y+2)$ b) $3a(b-4c+1)$ c) $-5(3+5t+mt)$ d) $4k(3m-5n+7)$.

4-masala

a) Tugallanmagan: qavsda hali $3p$ bor. To'g'risi: $9p(2q-3r)$.
b) $5x$ chiqqanda uchinchi had o'rnida 1 qolishi kerak: $5x(2y-3z+1)$.
c) Ishora xato: $-3\cdot(-4b)=+12b$. To'g'risi: $-3(3a+4b)$.

5-masala

a) $(2a+5)(3x-4y)$ b) ikkinchi hadda 1 qoladi: $(3p-2)(k+1)$ c) $(m-n)(5b+13b)=18b(m-n)$ d) $(c-d)(8a-12b)=4(c-d)(2a-3b)$.

6-masala

a) $x(x+7)$ b) $4ab(2a+3)$ c) $5mn(3m-2n)$ d) $p^2(p+4q-1)$ e) $-x^2y^2(y+x)$.

7-masala

a) $56\cdot100=5\,600$ b) $4{,}7\cdot1=4{,}7$ c) $100\cdot41=4\,100$ d) $19\cdot(-10)=-190$ e) $77\cdot100=7\,700$.

8-masala

a) $m(k+t)=13\cdot100=1\,300$.
b) $6p(k-3m)=24\cdot(35-27)=24\cdot8=192$.

9-masala

a) $15x + 10 - 3x + 18 = 12x + 28 = 4(3x + 7)$.
b) $20a - 4b + 2a - 6b = 22a - 10b = 2(11a - 5b)$.

10-masala

a) $-3b(y - 2x) = +3b(2x - y)$: $(2x - y)(4a + 3b)$.
b) $-3n(-m + 5n) = +3n(m - 5n)$: $(m - 5n) + 3n(m - 5n) = (m - 5n)(1 + 3n)$.

11-masala

a) $5x(y + 2) + 3(y + 2) = (y + 2)(5x + 3)$.
b) $4a(b + 3) - 7(b + 3) = (b + 3)(4a - 7)$.

12-masala

a) $8mn + 20m = 4m(2n + 5)$ — eni $2n + 5$.
b) Masalan: $7xy - 14x = 7x(y - 2)$ — tomonlari $7x$ va $y - 2$; yoki $7xy - 14x = x(7y - 14)$ — tomonlari x va $7y - 14$.

13-masala

EKUB$(15;\ 10;\ 25) = 5$. 5 ning 1 dan katta bo'luvchisi faqat 5 — demak yagona usul: 5 ta guruh, har birida $3x + 2y + 5$: $15x + 10y + 25 = 5(3x + 2y + 5)$.

14-masala

a) $28a + 42b$.
b) EKUB$(28;\ 42) = 14$ — ko'pi bilan 14 ta maktab, har biriga 2 quti daftar va 3 quti ruchka: $28a + 42b = 14(2a + 3b)$.

15-masala

a) $n(n + 3)$.
b) n juft bo'lsa, ko'paytma juft. n toq bo'lsa, $n + 3$ juft (toq + toq = juft) — ko'paytma yana juft.
c) $5k + 5k^2 = 5k(1 + k) = 5 \cdot k(k + 1)$; $k(k + 1)$ juft, demak $5 \cdot 2t = 10t$ — 10 ga bo'linadi.

16-masala

a) $3a(b+3)-6(b+3)=(b+3)(3a-6)=3(b+3)(a-2)$.
b) $a-2=0$, ya'ni $a=2$ bo'lsa, ifoda har doim 0.
c) $3\cdot(7+3)\cdot(5-2)=3\cdot10\cdot3=90$.

17-masala

a) $S_B-S_A=3x(4y+1)-5x(2y+3)=12xy+3x-10xy-15x=2xy-12x=2x(y-6)$.
b) $y=6$ da ayirma 0 — yuzlar teng. $y=10$ da $2x\cdot4=8x>0$ — B ning yuzi katta.

18-masala

a) $3(m-2n)$ c) $a(m-2n)$ d) $-5(m-2n)$ e) $m(m-2n)$. b) bo'lmaydi: $2m-n$ hadlarida umumiy ko'paytuvchi yo'q, u $m-2n$ ning o'zi ham emas. Masalan, yana: $4y(m-2n)=4my-8ny$.

5-qism · Tekshiramiz

1-masala

a) $7(a+2b)$ b) $6(4x-3)$ c) $3m(3n+2)$.

2-masala

a) $15p(3q-2r)$ b) $-4y(3+5x)$ c) $4a(2b-1+3c)$.

3-masala

Tugallanmagan: qavsdagi ikkala hadda z bor. To'g'risi: $7z(2x-3y)$.

4-masala

$(5a-c)(2t+9t)=(5a-c)(11t)=11t(5a-c)$.

5-masala

$9mn(2m+3n)$.

6-masala

a) $93\cdot100=9\,300$ b) $2{,}5\cdot4=10$.

7-masala

$12x - 6 - 4x + 12 = 8x + 6 = 2(4x + 3).$

8-masala

$12ab + 30a = 6a(2b + 5)$ — eni $2b + 5.$

9-masala

$-5b(4 - m) = +5b(m - 4)$: $(m - 4)(3a + 5b).$

10-masala

$6x(y + 3) + 5(y + 3) = (y + 3)(6x + 5).$